
Activation Steering and Representational Geometry in JEPA World Models

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Abstract

We study activation steering and representational geometry in frozen JEPA-WM checkpoints, comparing low-rank predictor corrections, visual and action-conditioning edits, and matched random controls. Layer sweeps, component ablations, and controlled replays measure how internal edits propagate into forecasts and planning. A learned rank-four correction reduces six-step proprioceptive forecast MSE by 2.36% on Reach and 2.19% on Reach-Wall. A score-margin bound certifies unchanged winners in 184 of 192 fixed candidate banks, while iterative search can still return different plans. Action-condition patches reproduce an input-action counterfactual only when both appearances in the context history are replaced. Under fixed weights and inputs, the comparison between cubic and linear activation reconstruction reverses between FP32 and BF16 at every predictor block. A 56-state physical replay and a separate 3,072-run protected evaluation assess the behavioral consequences; neither establishes improved physical performance under the registered simultaneous intervals. The study connects properties of activation edits to specific changes in prediction, scoring, and search, with control performance evaluated separately.

1 Introduction

We investigate activation steering inside frozen JEPA-WM checkpoints [3]. Our starting questions concern representation: which predictor layers respond to low-rank corrections, whether local interpolation provides useful steering directions, and how visual and action-conditioning edits interact. Because JEPA-WM scores candidate actions through its predictions, we can also follow these interventions into candidate rankings, adaptive search, and physical execution.

Joseph et al.’s Physics Emergence Zone study provides a useful reference [1]. Their layerwise probes, subspace analyses, and targeted interventions characterize distributed physical representations in video encoders. We examine an action-conditioned predictor and its planner. This changes both the intervention target and the relevant endpoints: a direction that changes an internal representation must also affect the relative scores of candidate plans to influence selection.

We compare a learned four-direction predictor correction with a response-calibrated random subspace, visual-action coupling with matched random directions, and joint versus single-component edits. Earlier layer and interpolation sweeps provide complementary measurements with their own fitted operators. Throughout, base weights and the planning objective remain fixed. Fitting, development diagnostics, and protected behavioral evaluation retain separate data populations.

The study follows layer responses and forecast corrections into fixed-bank selection, adaptive search, and physical execution. Action-history and numerical-precision controls test how to interpret the edits. A separate protected evaluation assesses behavioral utility. This organization connects internal effects with their downstream consequences while retaining the boundaries between experiments.

How JEPA-WM plans with predicted futures

Frozen MetaWorld model / alternative steering sites at imagined step H3

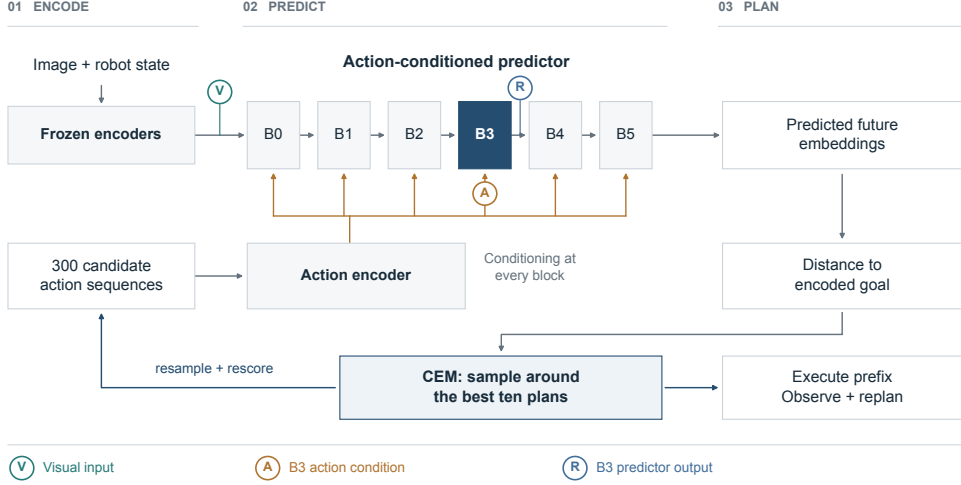


Figure 1: Steering inside a JEPA world model. The predictor forecasts the consequences of candidate action sequences. The planner scores their distance to the encoded goal, refits its action distribution, and repeats. It executes the beginning of the chosen plan before observing and replanning. R marks the four-direction predictor edit; V and A mark the alternative visual and action-conditioning edits. All edits occur at imagined step H3 of an H6 forecast. R is never combined with V/A in the eight-condition evaluation.

2 Related work

DINO-WM and JEPA-WM use prediction in learned features for planning [3, 5]. JEPA-WM studies which model and training choices support physical planning. We keep the released models fixed and ask what an inference-time activation correction changes.

Activation steering also appears in robot policies. COAST uses success- and failure-derived conceptors in a policy’s action expert [2]. Our intervention enters a world-model predictor whose output is scored by a separate sampling planner. The supervision and computations differ, so comparing reported gains does not isolate the reason for their difference.

Joseph et al. probe physical variables and intervene on motion representations in video models [1]. Their layer analyses motivate checking where an internal signal is accessible and what changes when it is edited. Our six-layer action-conditioned predictor is a different object of study, and its attention maps do not identify a physics-emergence zone. Curved activation interpolation [4] motivates a second control: reconstruction quality should be tested against downstream prediction and numerical precision before receiving a physical interpretation.

3 Model and experimental setup

Prediction and planning. Frozen encoders map images and robot state to feature vectors. A six-block predictor takes these features and candidate actions to forecast six imagined steps. In the MetaWorld tasks, the planner minimizes

$$C(a_{1:H}) = \text{MSE}(\hat{z}_H^v, z_g^v) + 0.1 \text{MSE}(\hat{z}_H^p, z_g^p), \quad (1)$$

where v and p denote visual and robot-state features, and each MSE averages over its feature dimensions. The target is the goal, whereas forecast error compares a prediction with the outcome of those actions.

The Cross-Entropy Method (CEM) starts with 300 action sequences, keeps the ten with lowest predicted cost, and samples a new population around them. MetaWorld uses fifteen such iterations per planning call. The robot executes a three-step prefix, corresponding to fifteen elementary sim-

ulator actions, before replanning. Full MetaWorld episodes contain seven planning calls over 100 elementary steps; navigation uses thirty search iterations per call.

Interventions. The main learned edit adds a normalized correction in a four-dimensional subspace to the fourth predictor block’s output (B3), at the third imagined step (H3). Its coefficients come from an offline fitted map of the current unedited features. A random-subspace comparison matches rank, edit site, and prescribed dose, with separate response calibration. Both are activation edits; the robot still obtains its actions from the planner.

We also test a learned pair of visual-input and action-conditioning directions. Equal-budget and randomized versions form one comparison. Unscaled joint, visual-only, and action-only edits form a component ablation. Together with unsteered and the two four-direction conditions, these give eight arms. Appendix A gives the equations, fitting details, and dose definitions. No intervention updates the base model weights.

Data and uncertainty. Fitting trajectories determine the operators. Separate recorded development trajectories measure prediction error and inform the recipe. Development diagnostics compare fixed plans, iterative searches, and short physical executions. The protected evaluation uses 96 new scenarios per task on Reach, Reach-Wall, PointMaze, and Wall after freezing the eight-arm recipe. All arms share the same physical GPU within each scenario, with strict FP32 and TF32 disabled. Engineering checks use excluded inputs.

Statistical units are source trajectories for recorded forecasts and starting scenarios for planning and behavior. Candidate plans are not independent replicates. Registered uncertainty uses paired bootstrap resampling and correction across the specified comparisons: six for the search extension, twelve for physical replay, and 48 for protected behavior. Each uses 20,000 draws. Details and source identities are retained in the appendices and accompanying reports. Push-T and DROID appear only in earlier development results; DROID measures recorded-action agreement.

4 Results

4.1 The learned edit improves some forecasts

On recorded development trajectories, the four-direction edit reduces H6 robot-state embedding MSE by 2.360% on Reach and 2.192% on Reach-Wall. These are prediction errors for recorded futures. They do not measure how the robot moves under the edited planner.

An earlier rank-one sweep finds larger H6 forecast corrections at early predictor blocks on both tasks, in FP32 and BF16. Push-T does not show the same pattern. The sweep uses independently fitted rank-one operators; it does not test moving the later four-direction edit to another layer. Appendix B retains the full layer analysis.

4.2 A score-margin bound characterizes fixed-bank selection

We next score identical banks of 300 action sequences in 192 development states, 96 per MetaWorld task. The learned edit changes the winning plan in zero Reach states and one Reach-Wall state.

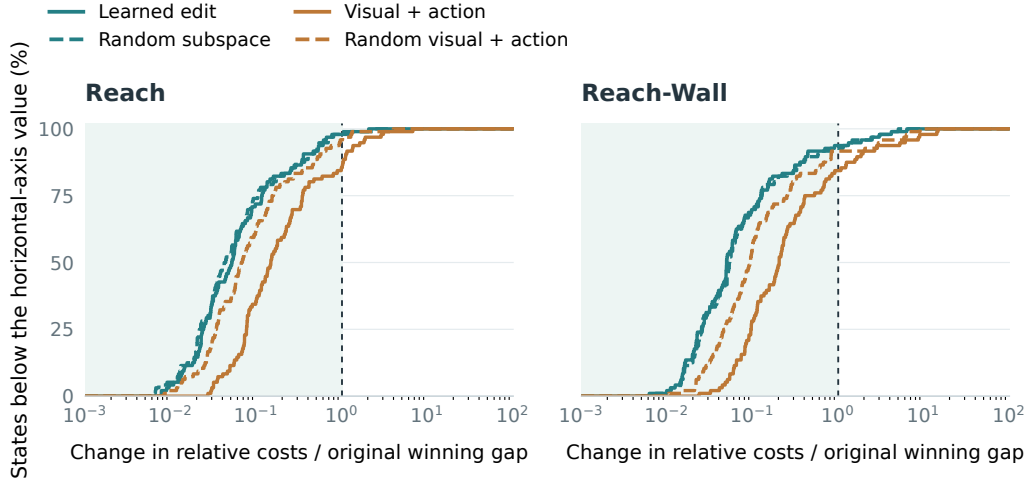
To assess why a winner can remain stable, let m be the original cost gap between the best and second-best plans. Let δ_i be the cost change for plan i , and let $r = \max_i \delta_i - \min_i \delta_i$ be the range of these changes. If $r < m$, no competitor can overtake the winner: its original disadvantage exceeds any relative improvement the edit can give it. Appendix C gives the proof and all four edit comparisons.

This condition holds for the learned edit in 184 of 192 states: 94 Reach and 90 Reach-Wall. Failure of the bound does not imply a changed winner. Conversely, preserving the best plan does not preserve all ten elites used by CEM. The bound applies to the saved finite set of candidates, and does not certify the result of later search iterations.

A separate 64-state replay asks which part of the edit produces the score changes. Keeping only the correction shared across candidates reconstructs the full edit’s relative cost changes with scores of 0.983 on Reach and 0.998 on Reach-Wall. Random edits show a similar pattern. A shared activation correction can still change relative costs because squared goal distance depends on each candidate’s

The first choice stays the same in 191 / 192 states

Learned edit on fixed candidate sets. The curves show all four edits.



Learned edit: 184/192 certified unchanged; 1/192 observed winner changes.

96 contexts/task · same 300 actions per comparison · initial population, not an entire CEM search

Figure 2: Most fixed candidate sets retain their winner. The horizontal axis divides the range of score changes by the original winner’s lead. Values below one certify an unchanged winner. Curves retain all 96 states per task and all four edits; the learned edit changes one winner across both tasks. This comparison holds candidate actions fixed.

starting prediction (Equation 6). Components retain their natural magnitudes, so this test does not compare their efficacy at equal energy.

4.3 Iterative search can return different plans

CEM changes its next candidate population using the current elite plans. Small score changes can therefore alter later searches even when the first winner remains unchanged. A 56-state extension, with 28 states per task, runs all fifteen iterations for unsteered, learned, and random-subspace conditions.

Both edits change the returned action prefixes. Mean RMS differences from unsteered are 0.697/0.578 for learned/random on Reach and 0.500/0.217 on Reach-Wall, in normalized action coordinates. All six registered learned-minus-random intervals include zero. These measurements establish changed plans, but do not establish an advantage for the learned edit or measure physical displacement. Complete search curves and contrasts appear in Appendix C.1.

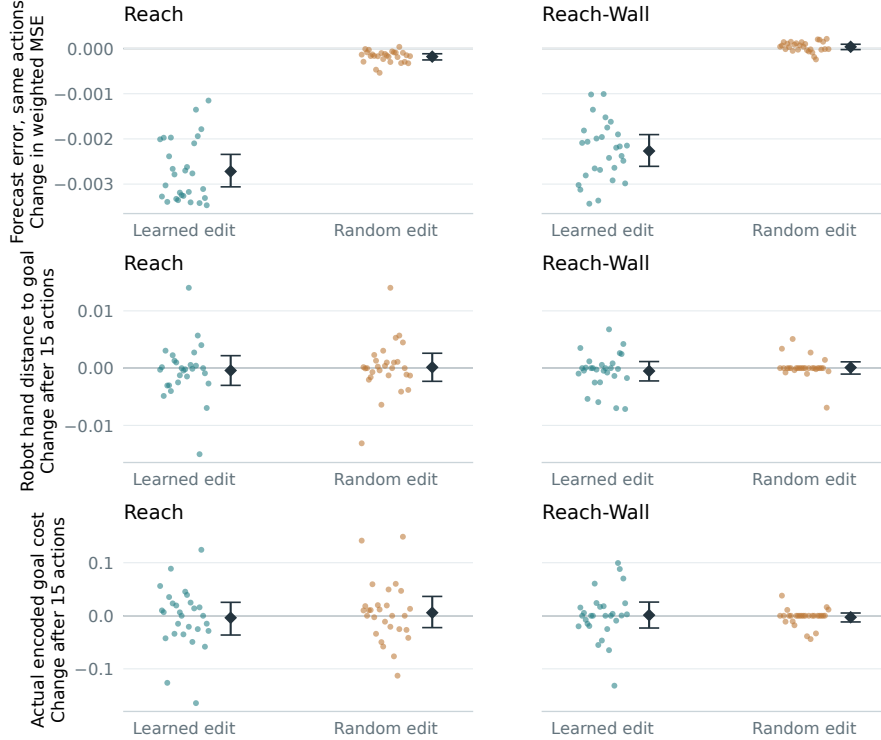
4.4 Physical replay separates forecast accuracy from plan quality

We execute each condition’s selected prefix from the same simulator reset in all 56 extension states. Each trajectory contains fifteen elementary actions. A repeated unsteered execution checks exact reset and trajectory agreement, giving 224 trajectories. All three models also predict all three selected plans, giving 504 model/plan forecasts.

Holding the unsteered-selected actions fixed, the learned edit reduces weighted forecast MSE by 0.002720 on Reach (simultaneous 95% interval [0.002342, 0.003061]) and 0.002268 on Reach-Wall ([0.001905, 0.002607]). These are absolute error reductions. For executed plans, however, all eight contrasts in physical end-effector distance and actual encoded-goal cost include zero under the registered twelve-comparison correction (Figure 3).

Better forecasts; no established physical gain

Same actions for prediction tests; each model's chosen actions for execution.



28 paired states per task. Bars: simultaneous 95% intervals across 12 comparisons.

Negative values favor the edit. Short executions on development states.

Figure 3: Forecast accuracy and physical outcome in the same 56 states. Negative effects favor the edit. The top row holds actions fixed and measures prediction error. The lower rows execute each model's selected actions and measure the resulting distance and goal cost. Dots show paired states, diamonds show means, and bars show the registered family-twelve intervals. Learned forecasts improve in both tasks; all eight physical-outcome intervals cross zero.

Thus we can detect better same-action forecasts in this experiment without establishing that the selected actions bring the robot closer to its goal. This tests short executions on development states. It is separate from full-task success. CEM optimizes H6 while this follow-up measures the H3 endpoint, so discrepancies here do not by themselves show exploitation of the optimized objective.

4.5 Action history and precision constrain the interpretation of an edit

The action-conditioned predictor provides a controlled test of intervention semantics. An H3 action appears twice in its two-frame context: newest at H3, then older at H4. Across sixteen development contexts and two candidate banks, replacing the action-derived condition at both appearances and all six blocks reproduces the raw input-action substitution exactly through H6. A one-time patch matches at H3 but diverges when the original action returns as history. Appendix C.4 retains the full layer contrasts and input-range controls.

We also test local activation reconstruction along fixed action perturbations. Holding weights, inputs, and interpolation coefficients fixed across 64 contexts, cubic reconstruction outperforms linear reconstruction at every block in FP32; BF16 reverses the comparison. Rounding only the FP32 output moves the two estimators toward equality but does not reproduce actual BF16 computation. A separate five-task study finds small, mixed downstream forecast benefits after matching requested edit

Table 1: Complete protected success rates (%). Each cell has 96 paired scenarios. All 48 registered simultaneous contrast intervals include zero.

Arm	Reach	Reach-Wall	PointMaze	Wall
Native	54.17	29.17	86.46	81.25
Refined rank-four	54.17	20.83	88.54	80.21
Calibrated random subspace	44.79	21.88	86.46	80.21
Equal-budget coupling	47.92	18.75	84.38	83.33
Dose-matched random directions	46.88	27.08	87.50	77.08
Unscaled joint	52.08	22.92	84.38	77.08
Visual only	54.17	22.92	81.25	80.21
Action conditioning only	47.92	17.71	81.25	78.13

doses. These results distinguish local interpolation quality from the utility of the resulting correction. Appendix D gives the numerical controls and complete measurements.

4.6 Protected behavioral evaluation

The protected panel evaluates eight arms on 384 fresh scenarios across four tasks, for 3,072 runs. Table 1 gives every arm. All 48 registered simultaneous intervals include zero, covering comparisons with unsteered, learned versus random edits, and component interactions. This is insufficient evidence for a reliable success gain; it is not an equivalence result.

Steering does change which scenarios succeed. On Reach, unsteered and learned steering each succeed on 52 of 96 scenarios. The edit rescues 21 failures and loses 21 successes. Earlier development rates use different scenarios and planner randomness, and are not substitutes for the concurrent protected baseline.

5 Discussion and limitations

For a JEPA planner, the practical test of a prediction correction is whether it helps select actions with better outcomes. Our results show that forecast error, candidate ordering, search behavior, and physical performance can respond differently to the same intervention family. Checking only the first measurement leaves the planning claim unresolved.

The fixed-bank bound gives a precise explanation for many unchanged initial choices. It does not explain the protected success result: those episodes used different inputs and random seeds, and did not retain numerical candidate costs or elite identities. The physical follow-up measures three selected plans per state, not the true physical ranking of all 300 candidates. We therefore cannot conclude that a particular ranking error caused the lack of a success gain.

The study uses one released checkpoint per task, task-specific fitted edits, and 96 protected scenarios per task. It has no independent training-seed replication and leaves small behavioral effects unresolved. Development diagnostics informed the analysis and should not be read as additional protected confirmation. Full source tables retain negative results, random controls, and unavailable historical cells.

6 Conclusion

The experiments characterize activation steering in a frozen JEPA world model through layer responses, component effects, action-history consistency, and numerical controls. A score-margin bound links cost perturbations to fixed-bank selection; adaptive-search and simulator replays extend the measurement to returned plans and physical endpoints. These interventions improve some forecasts and alter some plans, while the protected evaluation leaves behavioral gains unestablished. The distinction makes both the internal effects and their practical limits explicit.

References

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Table 2: Eight arms, three alternative intervention sites. Native plus these seven edits form the protected panel. Earlier layer and geometry sweeps are separate. R is never combined with V/A.

Family	Intervention	Comparator / ablation
Refined	Four-direction R at B3 output	Calibrated random subspace
Equal-budget coupling	V+A; each dose scaled by $1/\sqrt{2}$	Same-dose random V+A
Component ablation	Original-dose V+A	V only; A only

A Intervention details and data populations

Frozen predictor and planner. We use released JEPA-WM checkpoints and task-specific fitted interventions. Reach and Reach-Wall share a checkpoint, not their fitted operators. DINO visual features and proprioceptive features feed a six-block action-conditioned predictor. All base encoders and predictor weights remain frozen during intervention; this does not imply that every encoder was frozen during original training.

The analyzed MetaWorld planner uses the actual weighted goal cost

$$C(a_{1:H}) = \text{MSE}(z_H^v, z_g^v) + 0.1 \text{MSE}(z_H^p, z_g^p), \quad (2)$$

with each MSE averaged over its modality’s dimensions. The cost is *not* visual-only. Recorded-future proprioceptive error and goal-directed planning cost nevertheless use different targets, weights, and action inputs. CEM uses 300 candidates. MetaWorld uses 15 optimization iterations and seven planning calls over the recorded 100-elementary-step episodes; navigation uses 30 iterations. Imagined steps, simulator steps, CEM iterations, and replanning calls are distinct units.

Refined rank-four correction. The later fixed-response operator edits the B3 output at H3. With native field h , orthonormal four-direction basis B , and $\phi(h)$ consisting of an intercept and three standardized native-feature scores, an offline map W gives

$$d(h) = BW\phi(h), \quad h' = h + \eta d(h)/\|d(h)\|_2, \quad (3)$$

using the frozen zero-edit rule below the correction threshold. The edit covers all 256 visual patches within the existing batch, without online response probes or native-shadow forecasts. Its random-subspace comparator has separate response calibration and matched support, rank, basis-projector spectrum, and prescribed dose—not identical coefficient-map singular values. This successor is not an equivalent implementation of the earlier online response solver.

Visual/action interventions. A fit-only maximum-covariance direction pair edits the H3 visual predictor input and B3 action-conditioning vector. Unscaled joint, visual-only, and action-only arms form a fixed-dose factorial. Equal-budget coupling scales both standardized doses by $1/\sqrt{2}$; its random-direction comparator uses the same sites and dose policy. This comparison is distinct from the unscaled factorial (Table 2).

Separate populations and units. Fit trajectories supply directions, readouts, and calibration. Separate recorded-action development lineages measure forecast effects and inform choices; fitting-disjoint does not mean selection-protected. Corrected primary offline pools contain 33 Reach, 27 Reach-Wall, and 21 Push-T lineages. Geometry analyses also include 192 Wall and 200 PointMaze lineages. The archived decision diagnostic has 96 canonical development scenarios per MetaWorld task, each with one shared initial bank of 300 action sequences. Protected confirmation instead uses 96 *new* scientific scenarios per task on Reach, Reach-Wall, PointMaze, and Wall. The earlier six-task behavioral table and interrupted native-only cohort remain separate.

The archival CPU analyses recount hash-verified records. New GPU diagnostics separately reexecute the frozen predictor on 64 canonical development contexts (IDs 0–31 per MetaWorld task), with an eight-context actual-CEM pilot (IDs 0–3 per task). Controlled arithmetic uses all 64 contexts; action-conditioning specificity uses 16 (IDs 0–7 per task), each with original and freshly seeded candidate banks. Fresh candidates are not fresh scenarios. These diagnostics do not execute physical actions. Offline effects use source lineages as units; candidate costs and behavior use scenarios. Candidates, heads, horizons, and calls are not independent replicates. Follow-up protocols specify their uncertainty families before outcomes; the studies remain development analyses, not replacements for protected confirmation.

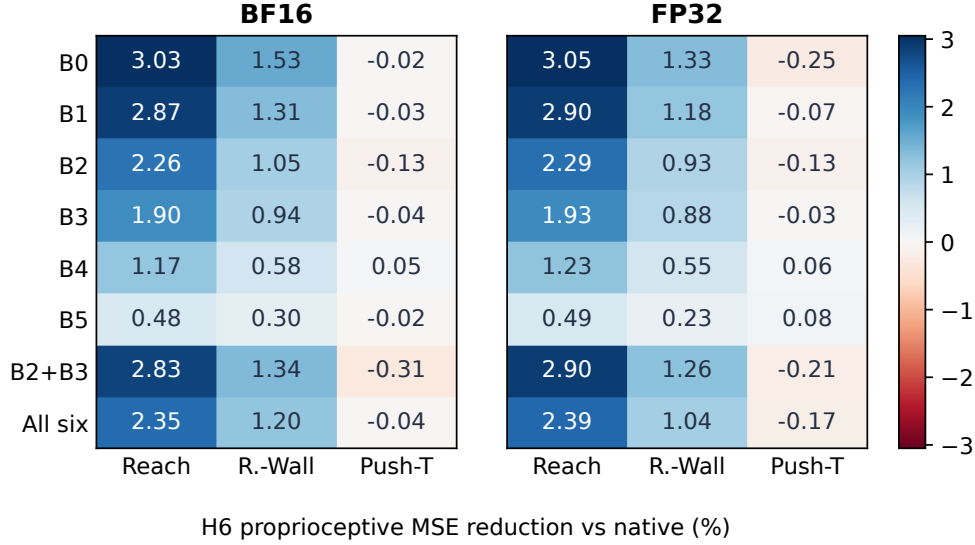


Figure 4: An early-to-late forecast response in the rank-one sweep. H6 proprioceptive-error reduction relative to native in BF16 and FP32, retaining all eight intervention supports and three tasks. The full 576-cell support/modality/horizon grid and matched-random comparisons are provided separately. These are independently fitted rank-one interventions, not a layer sweep of the later rank-four successor.

B Layer-by-layer forecast effects

A recount verifies all 576 task/precision/support/modality/horizon cells in the earlier layer grid. In its singleton *rank-one* sweep, H6 proprioceptive-error reduction decreases monotonically from B0 to B5 on both MetaWorld tasks in both numerical conditions (Figure 4). Reach B0/B5 reductions are 3.026%/0.475% in BF16 and 3.050%/0.494% in FP32; Reach-Wall reductions are 1.526%/0.298% and 1.334%/0.228%. Original BF16 simultaneous B0-minus-B5 intervals are [1.769, 3.333] and [0.243, 2.214] percentage points of native MSE.

This is intervention susceptibility, not a uniquely semantic layer. Layer-specific fits, representation scale, dose realization, and downstream computation all enter the comparison. Reach B0 versus B1 remains unresolved, and Push-T does not reproduce the MetaWorld H6 benefit. Endpoints before H3 precede the intervention.

The later *rank-four fixed-response* successor is evaluated at B3, not across this layer sweep. It reduces development H6 proprioceptive MSE by 2.360% on Reach and 2.192% on Reach-Wall while removing online response probes. Neither result establishes that moving the successor to B0 would improve planning.

Table 3: All diagnostic arms, 96 banks per task. Certified: $r < m$. Changed: actual winning-index changes. RMS is candidate-centered cost change divided by native within-bank cost standard deviation, then averaged over scenarios.

Task	Arm	Certified	Changed	RMS (%)
Reach	Refined	94	0	0.2466
	Random subspace	95	0	0.2116
	Coupling	82	1	0.7625
	Random directions	92	0	0.3461
Reach-Wall	Refined	90	1	0.2786
	Random subspace	89	0	0.2922
	Coupling	81	2	1.0786
	Random directions	88	1	0.4748

C Candidate scores, component replay, and search

An exact sufficient certificate. For a fixed candidate bank, let C_i be native cost, i^* its unique winner, and $\delta_i = C'_i - C_i$ the edit-induced change. Define

$$m = \min_{j \neq i^*} (C_j - C_{i^*}), \quad r = \max_i \delta_i - \min_i \delta_i. \quad (4)$$

If $r < m$, the winner cannot change, because every competitor satisfies

$$C'_j - C'_{i^*} = (C_j - C_{i^*}) + (\delta_j - \delta_{i^*}) \geq m - r > 0. \quad (5)$$

This sufficient inequality concerns the saved finite bank, not generalization. Failure to certify does not imply a flip; actual selection and tie handling are checked separately. The diagnostic selects the bank’s best candidate. CEM instead updates its proposal from ten elites, so preserving the best index alone does not certify an unchanged CEM update or final plan.

Measured cost changes are small relative to ranking. We analyze all 192 completed development scenarios and four non-native diagnostic arms (768 paired comparisons). For refined editing, $r < m$ in 94/96 Reach and 90/96 Reach-Wall banks; winners actually change in 0/96 and 1/96. The mean scenario-level ratio

$$100 \text{ RMS}_i(\delta_i - \bar{\delta}) / \sqrt{\text{Var}_i(C_i)}$$

is 0.2466% and 0.2786%. Coupling certifies 82/96 and 81/96 banks and changes 1/96 and 2/96 winners. Table 3 retains both random controls.

Shared coefficients need not produce shared costs. A separate decomposition of 768 protected arm records finds that 98.855% and 99.615% of refined coefficient energy on Reach and Reach-Wall is shared across candidates within a call. This is coefficient geometry, not the fraction of decision-irrelevant information. Even a constant final-latent translation d changes squared goal cost by

$$\|z_i + d - g\|^2 - \|z_i - g\|^2 = 2d^\top(z_i - g) + \|d\|^2, \quad (6)$$

whose first term varies across candidates. The actual edit also passes through a nonlinear predictor. Margin measurements therefore add information that coefficient energy alone cannot supply.

Archived decision banks and protected episodes differ in inputs and planner RNG. Protected records did not retain numeric candidate costs or elite identities. Thus the certificate cannot explain which protected scenarios were rescued or broken. A low first-bank flip count also does not establish replanning drift as their cause.

C.1 Component replay and actual adaptive search

A shared field carries relative-score changes. We test the shared-correction question on all 64 fixed development contexts. For each learned or calibrated-random correction, cache its native H3/B3 field and replay the full field, zero field, candidate-common mean, and candidate-centered remainder on the same 300 actions. Cached-full and zero controls reproduce their reference forecasts and scores exactly. Neither component is renormalized, and no operator is refitted.

Small forecast edits. Different search paths.

Same initial candidate actions. Separate adaptive searches.

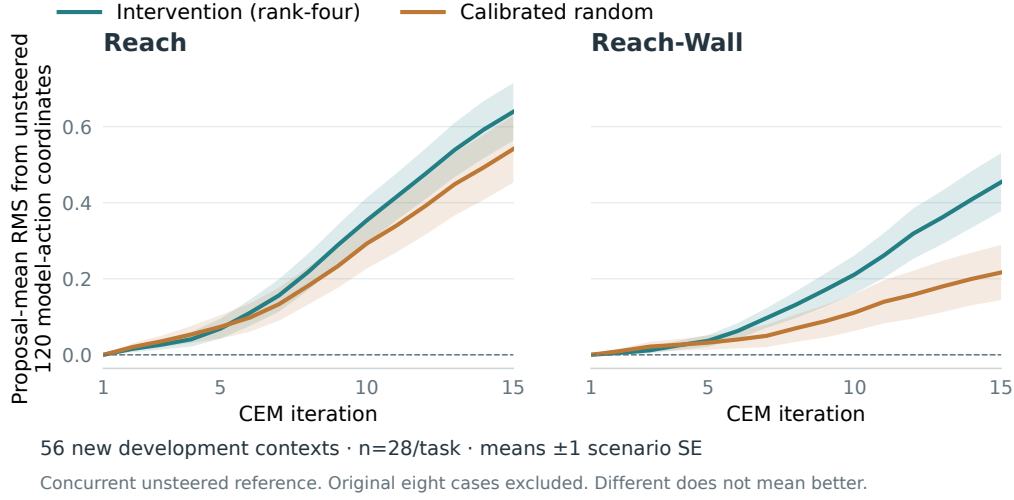


Figure 5: Adaptive search can diverge despite nearly shared initial elites. Complete 56-context extension, both edit arms and all fifteen iterations. Shading is one scenario standard error, not a confidence interval. RMS uses all 120 proposal-mean coordinates; the returned prefix uses 60 coordinates. Entropy curves and every paired returned-prefix endpoint are retained in the public CEM report. This is a search measurement, not a physical-quality test.

Let u be the full intervention’s cost change, centered across candidates, and u_k the corresponding change for component k . We report $R_k = 1 - \|u - u_k\|_2^2 / \|u\|_2^2$, retaining undefined zero-denominator cases and allowing negative values. This is a reconstruction score, not a probability. For learned edits, common-only R is 0.9827 on Reach (marginal 95% interval [0.9793, 0.9858]) and 0.9975 on Reach-Wall ([0.9971, 0.9980]); centered-only values are 0.0218 and 0.0135. Common-field energy fractions are 99.53% and 99.78%. Random-subspace common-only scores are similarly high, 0.9795 and 0.9854. These results establish the common component’s effect at its natural energy, not superiority over equal-energy centered steering.

Small initial changes can accompany divergent adaptive searches. After an initial eight-case pilot, a separately frozen extension reruns all three arms on the same receiving GPU for 56 new development contexts ($n = 28$ per task). Each search uses fifteen CEM iterations, 300 candidates and ten elites; traced/untraced selected-plan and RNG parity pass for all 336 searches. The original eight cases remain separate from primary inference. Every arm starts from identical candidate actions, but initial elite sets need not be identical: mean learned/random overlap with the native top ten is 9.893/9.857 on Reach and 9.964/9.929 on Reach-Wall.

Later proposals adapt to each arm’s scores, so later candidate IDs are not matched actions. Figure 5 retains all fifteen iterations. Returned three-step, 60-coordinate prefixes have mean RMS differences from unsteered of 0.697/0.578 for learned/random edits on Reach and 0.500/0.217 on Reach-Wall. These normalized model-action differences are not physical displacements or success gains. All six prespecified learned-minus-random intervals include zero (Table 4); the positive marginal Reach-Wall prefix contrast does not survive the registered family correction. Search-path changes therefore extend beyond the original eight cases, without establishing greater divergence or a consistent entropy effect for learned edits over calibrated random edits.

C.2 Forecast improvement and physical prefix quality remain distinct

The separately frozen physical follow-up covers all 56 extension contexts, $n = 28$ per task. Each unsteered, learned, and calibrated-random selected prefix executes from the original seeded simulator

Table 4: All six registered adaptive-search contrasts. Learned minus calibrated random, relative to concurrent unsteered; $n = 28$ paired contexts per task. Intervals use 20,000 paired-context bootstrap draws and Bonferroni family six. Proposal and entropy effects average all fifteen iterations.

Task	Metric	Mean	Simultaneous 95% interval
Reach	Prefix RMS	0.1197	$[-0.1284, 0.3487]$
	Proposal RMS	0.0403	$[-0.0717, 0.1353]$
	Entropy (nats)	0.2631	$[-0.8152, 1.4570]$
Reach-Wall	Prefix RMS	0.2827	$[-0.0070, 0.5294]$
	Proposal RMS	0.0809	$[-0.0541, 0.1929]$
	Entropy (nats)	0.3596	$[-0.6327, 1.3388]$

reset for fifteen elementary actions. A repeated unsteered trajectory checks byte-identical observations and physics, giving 224 trajectories in total. Every model forecasts every plan, retaining 504 scientific model/plan evaluations. Exact reset, native-repeat and H3 future-tail receiving checks pass; the preprocessing amendment and failed first attempt remain preserved in the accompanying physical-prefix report.

On the same unsteered-selected actions, learned-minus-unsteered weighted forecast MSE is -0.002720 on Reach (simultaneous 95% interval $[-0.003061, -0.002342]$) and -0.002268 on Reach-Wall ($[-0.002607, -0.001905]$). These are absolute errors, not percentages. All eight physical end-effector-distance and encoded-goal-cost contrasts for the two edit arms include zero under the prespecified twelve-contrast family correction (Figure 3). Thus this experiment establishes improved same-action forecasts in both tasks without an established physical-prefix gain. CEM optimized H6, whereas execution and scoring here cover H3; predicted/actual H3 preference disagreement alone does not establish exploitation of the H6 objective. This is development-prefix execution, not full-task success or another protected evaluation.

C.3 Action conditions affect rankings and elite membership differently

In a separate 16-context experiment, each candidate receives the next candidate’s newest action-derived AdaLN condition at H3, one block at a time. A private-generator isotropic perturbation matches each donor-minus-recipient L2 norm. All six blocks and both original and freshly seeded 300-action banks are retained. This is donor interchange within a context, not an intervention on explicit action tokens or semantic physical labels.

At B1 and B4, donor substitution disrupts native official-cost rankings more than the random edit: excess Spearman loss is 0.0071 – 0.0084 across the two tasks and two banks. Six-layer simultaneous intervals exclude zero in these cells, but no layer reaches the prespecified 0.05 practical flag. Native rank agreement remains at least 0.98979 in every official-cost cell. At B1, however, donor edits lose a mean 1.0 – 1.125 additional members of the native top ten relative to random edits. Small global rank changes and a changed elite set can coexist; neither establishes physical action quality.

Delivered-norm mismatch is not supported here: the maximum relative error is 2.72×10^{-8} . Distributional mismatch remains. The affine action encoder maps 20 input coordinates into 400 condition coordinates; isotropic noise need not lie in its at-most-20-dimensional reachable subspace. Matching total L2 therefore does not match action-reachable dose. Complete specificity curves and uncertainty appear in Appendix F; the following separately frozen study adds range controls.

C.4 Coherent action changes require consistent history

An H3 action appears twice in the predictor’s two-frame context: newest at H3, then older at H4. On the same 16 contexts and both fixed banks, we compare coherent raw H3 action substitution with donor-condition replacement at H3 only or at both appearances. All-block persistent replacement reproduces the raw-action counterfactual byte-exactly through H6. All-block H3-only replacement matches at H3 but has H6 reconstruction $R = 0.5029$ – 0.5079 across task/bank cells (Figure 7). Here $R = 1 - \text{MSE}(\hat{z}_{\text{patch}}, \hat{z}_{\text{cf}}) / \text{MSE}(\hat{z}_{\text{native}}, \hat{z}_{\text{cf}})$, using ratios of candidate means under the official modality weighting. This is a predictor-internal counterfactual, not its physical correctness.

Predicted endpoint versus the endpoint reached

Same three plans, evaluated by each model and executed after identical resets.

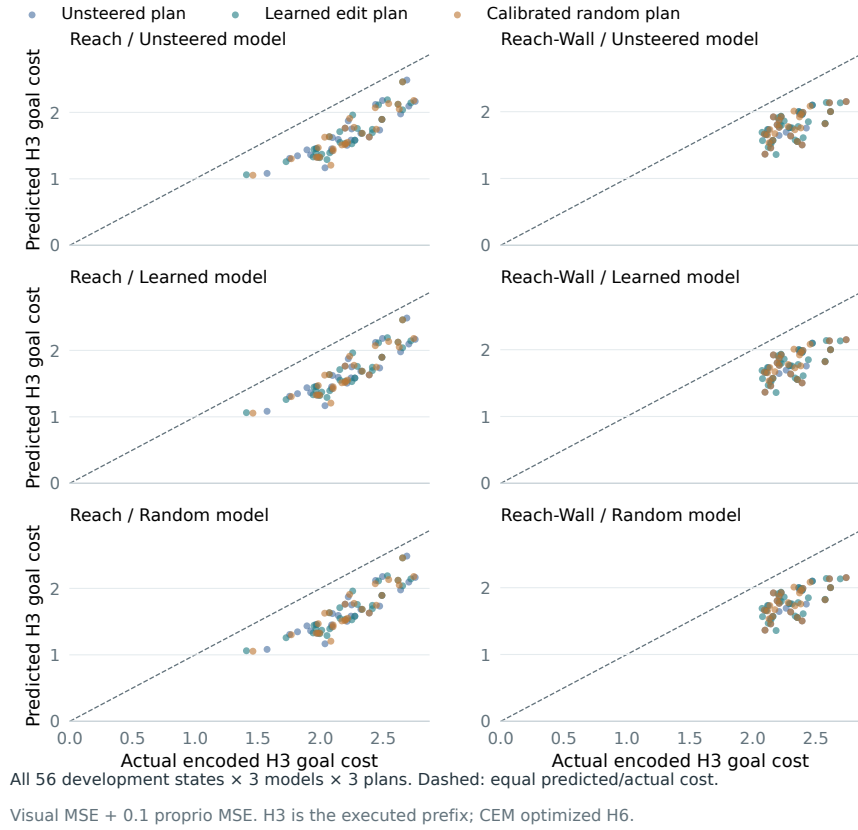


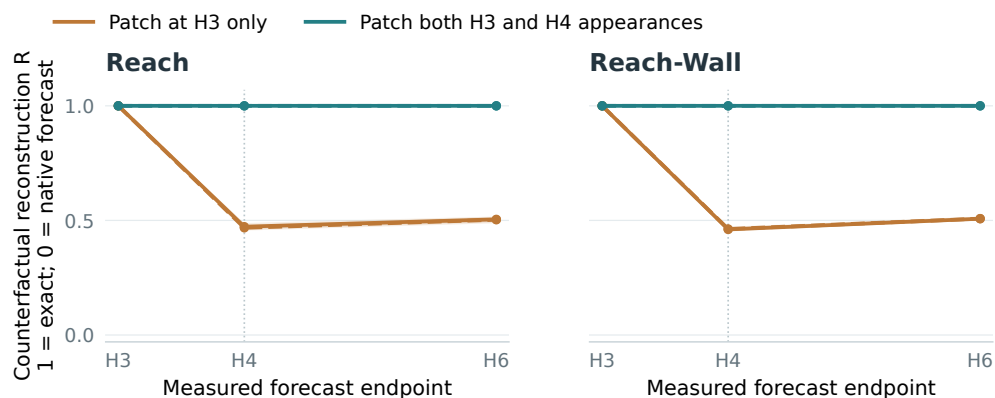
Figure 6: The complete prediction/execution grid. All three models predict all three selected plans in every context: 504 points. Axes use visual goal MSE plus 0.1 times proprioceptive goal MSE, with equal scales within task. Dashed lines mark equal predicted and actual encoded H3 goal cost. These are endpoint goal costs, distinct from prediction-versus-endpoint forecast error.

The six-layer contrasts retain a 72-cell simultaneous primary family (two tasks, two banks, six layers, three contrasts). Persistent versus H3-only reconstruction is positively separated at B0–B3 in both tasks/banks, 16 of 24 cells; B4/B5 remain unresolved. The largest observed persistence difference occurs at B1, $\Delta R = 0.4991$ – 0.5057 . H3 donor replacement exceeds equal-norm range-random reconstruction at B0/B1/B2/B4, but not established at B3/B5. These are a complete depth map, not a retrospectively selected intervention site.

Range-random edits disrupt native rankings more than off-range random edits at B1/B4 in both banks, by about 0.0021–0.0025 Spearman units; only fresh-bank B2 adds positive separation. The frozen encoder has rank 20 of 400; isotropic controls place 4.995% mean energy in its range. Actual norm error is at most 3.41×10^{-8} . Input-range membership is not a global activation manifold, and off-range does not mean causally inert. Together, these interventions identify an action-history dependency within this predictor. They neither reproduce the older static steering operators nor establish the cause of protected outcome changes.

An action returns in the next context window.

A one-time patch is not the same as changing that action.



All six blocks patched. The donor action reappears as history at H4.

Solid / dashed: original / fresh action banks · all 16 contexts, $n=8/\text{task}$ · full marginal 95% bands

Predictor-internal action counterfactuals, not physical outcomes or protected-confirmation mechanisms.

Figure 7: A one-time internal patch is not a coherent action change. The donor H3 condition must also replace the older occurrence at H4. All six blocks, both original/fresh action banks, and all 16 contexts are retained. The exact persistent/raw-action match is a computational consistency control; the H3-only deviation is measured. Bands are full marginal context-bootstrap intervals. No physical actions are executed.

Precision changes what looks locally curved

Same weights, same action perturbations, every predictor block.

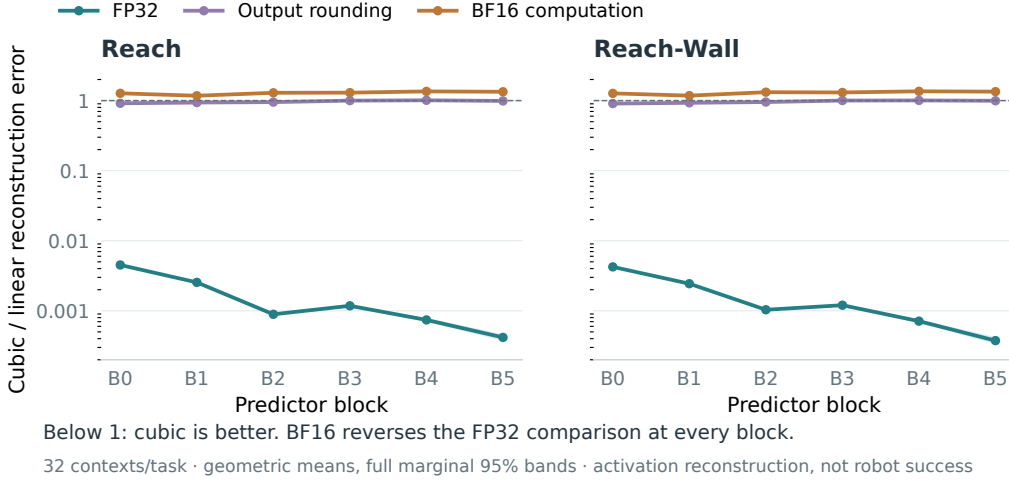


Figure 8: The local reconstruction comparison depends on computation, not only output rounding. Same weights, inputs, action perturbations and coefficients; 32 contexts per task, all six blocks. Curves show geometric mean cubic/linear activation-MSE ratios with full marginal 95% bands. The prespecified paired arithmetic contrasts use six-layer correction. These are reconstruction errors, not forecast or robot-success gains.

D Numerical precision and activation reconstruction

A controlled arithmetic comparison. The 64-context experiment fixes model weights, encoded inputs, the literal first action coordinate, four symmetric perturbations ($\pm 0.1, \pm 0.05$), and reconstruction coefficients. At every block, it compares strict FP32, actual BF16-autocast computation, and FP32 fields rounded to BF16 then restored to FP32. Each condition reconstructs its own unperturbed center; no fitted bank or dose-normalized correction changes between conditions.

FP32 favors cubic reconstruction at all six blocks: mean log cubic/linear MSE ratios span -5.40 to -7.89 . Output round-tripping moves these ratios near equality (-0.098 to $+0.010$), whereas actual BF16 produces $+0.160$ to $+0.306$. The actual-BF16-minus-rounded contrast is $+0.224$ to $+0.336$ across all twelve task/block cells, with six-layer-corrected lower bounds above the prespecified 0.1 log-unit margin. Thus output rounding degrades the apparent reconstruction advantage but does not quantitatively reproduce the actual BF16 result. The experiment isolates numerical-condition sensitivity, not an individual kernel or a physical manifold.

Precision-sensitive local interpolation. The action-response experiment withholds the center of a four-donor activation path. Coordinates $-1, -\frac{1}{2}, \frac{1}{2}, 1$ index frozen local action offsets. Equal-anchor linear reconstruction averages the donors. The archived cubic estimator at zero is

$$\hat{h}_{\text{center}} = -\frac{1}{6}h_{-1} + \frac{2}{3}h_{-1/2} + \frac{2}{3}h_{1/2} - \frac{1}{6}h_1. \quad (7)$$

At this symmetric evaluation point the same functional is obtained by quadratic least squares. It does not uniquely identify third-order dynamics or a dense activation manifold.

Across all five tasks, raw omitted-activation MSE is 35.15 – 46.77% *higher* than linear reconstruction in BF16, but 93.96 – 99.99% *lower* in FP32. This does not correspond to a similar improvement in future prediction. Under matched requested doses, the H6 proprioceptive-MSE advantage over the linear arm is small and mixed: BF16 task means range from -0.0213% to $+0.0264\%$ of native future MSE; FP32 ranges from -0.0141% to $+0.0051\%$. The two endpoints have different targets and denominators (Appendix H), so these scales are not an attenuation coefficient.

Requested dose is not exact realized-energy equivalence. The precision conditions also differ in fitted banks and prescribed dose, preventing attribution solely to rounding. In FP32 Reach/Reach-

Wall, the cubic arm’s per-window realized-energy tolerance passes only 49.0%/50.7% of windows, versus 98.1%/98.4% for linear reconstruction, despite small mean realized/requested L2 discrepancies. The comparison is therefore labelled by its actual requested-dose contract, not perfect per-window energy matching.

Output nonadditivity differs from loss interaction. For the unscaled visual/action factorial, let $e = z_0 - y$, $v = z_V - z_0$, $a = z_A - z_0$, and $q = z_{VA} - z_V - z_A + z_0$. Squared-error interaction obeys

$$L(VA) - L(V) - L(A) + L(0) = 2v^\top a + 2(e + v + a)^\top q + \|q\|^2, \quad (8)$$

with the endpoint’s dimensional averaging. Thus loss interaction can be nonzero even when output interaction q vanishes.

Saved H6 output-interaction squared norms, divided by native proprioceptive MSE, are 0.419%, 0.512%, and 5.020% in BF16 for Reach, Reach-Wall, and Push-T. Corresponding FP32 values are $4.82 \times 10^{-6}\%$, $1.07 \times 10^{-5}\%$, and 0.00256%. These establish precision-conditioned differences in retained scalar diagnostics, not that rounding caused them. Missing horizons remain missing; original tensors are not reconstructed from scalars. Together, these results distinguish local interpolation quality, the delivered activation intervention, and its downstream forecast utility. The loss terms can nearly cancel: BF16 Push-T has a +2.460% additive cross-term and -2.509% nonadditive remainder, yielding only -0.049% net H6 loss interaction. Figure 11 retains all seven source conditions, not just this illustrative case.

E Native attention and diagnostic evidence

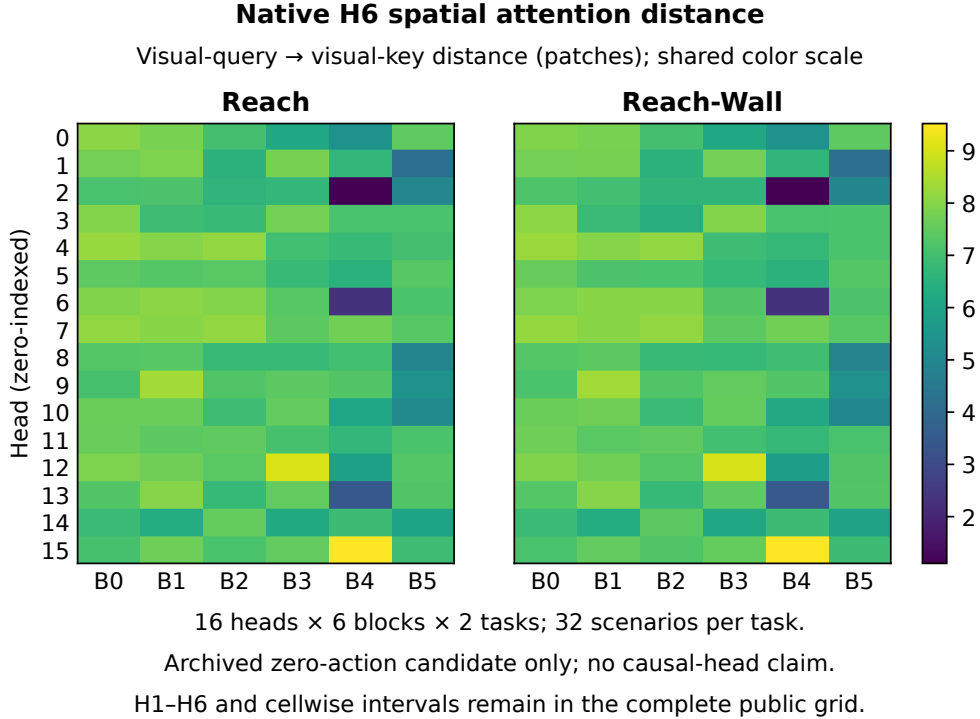


Figure 9: Native H6 attention distance, all six blocks and sixteen heads. Each cell averages 32 development contexts per task with the same archived zero-action candidate. Colors are shared across tasks. The complete H1–H6 grid is retained in the accompanying data. Attention distance is not causal importance or evidence of a planning-emergence zone.

The observer measures post-RoPE attention-weighted spatial and temporal distance while retaining the native attention kernel and exact output/RNG parity. It covers six blocks, sixteen heads and six

horizons on all 64 contexts. Actions enter through AdaLN conditioning rather than extra attention tokens; zero action-token mass is not evidence that actions are ignored. The zero-action input does not characterize attention over arbitrary candidate plans.

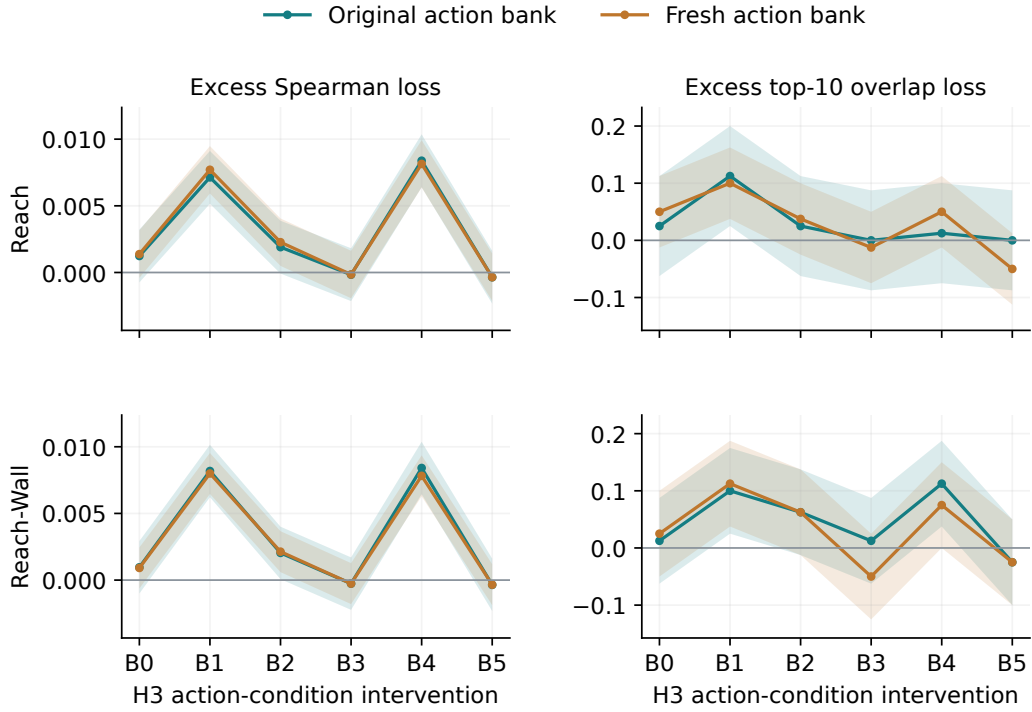
Native and steered CEM traces retain actual proposal standard deviations. Differential entropy is $\sum_{j=1}^{120} (\log \sigma_j + \frac{1}{2} \log(2\pi e))$, before clipping and insertion of the mean candidate. It depends on action units and may be negative. It is neither attention entropy nor calibrated uncertainty about physical outcomes.

All files emitted by these completed diagnostics are cloud-verified. Full H6 tensors were retained for the first eight contexts; later contexts retain compact diagnostics after on-GPU parity checks, under the fixed retention policy. Preservation does not recover tensors that were never saved. The eight actual-CEM follow-up records are fully preserved. An engineering-only amendment changed an incorrect fifteen-callback guard to thirty: each iteration forecasts both the 300-candidate population and its updated mean. Failed attempts remain excluded, with unchanged seeds and scientific arms.

F Action-conditioning specificity

The subsequent action-counterfactual experiment retains all 72 primary contrasts in the accompanying `action_counterfactual_primary_contrasts.csv`. Its 20,000 paired bootstrap draws share context weights across banks and layers. Each task/contrast family uses the maximum absolute centered deviation over twelve bank/layer cells, with quantile $1 - 0.05/6$ across six families. Undefined cells would remain undefined; all registered cells were defined. All 1,120 forecasts, source identities, cloud receipts, norm controls and full-horizon parity checks passed. The full-layer figure and reproduction receipt are provided alongside the focused timing figure.

Action alignment versus norm-matched random edits



Positive: permutation disrupts native rankings more than random.

All 16 cases; $n=8/\text{task}$. Six-layer simultaneous 95% bands.

Official visual + 0.1 proprio goal cost. No actions executed.

Figure 10: Small global ranking changes can alter the elite set. Complete 16-context donor-interchange experiment, eight contexts per task, two fixed candidate banks, and all six layers. The contrast is excess loss of native goal-cost rank agreement relative to an equal-norm isotropic edit. Six-layer simultaneous intervals are within task/bank/metric families, not across every secondary endpoint. No layer reaches the prespecified 0.05 excess-Spearman practical flag. Rankings are model predictions, not physical counterfactual values.

Zero-capture passes reproduce H1–H6 forecasts and costs exactly; interventions leave the earlier condition time unchanged. All 115,200 delivered norms pass. The 448 total forecasts include 32 engineering passes and do not create 448 independent observations. An import-only receiving amendment initializes the native planner dependency before the RNG boundary; the final guard still requires unchanged global RNG throughout all scientific forecasts. The failed attempt remains archived. The public source-bound receipt exports all 1,152 case metric rows and 144 contrast rows.

G Historical evaluation and published context

Table 5: Historical development only, not pooled into protected confirmation. DROID is a recorded-plan action score over 64 endpoints, not 96 physical robot episodes. MetaWorld component rows marked * use concurrent native 45.83/29.17, not the canonical native row.

Arm	Reach	R.-Wall	Push-T	P.Maze	Wall	DROID
Native	44.79	30.21	59.38	80.21	76.04	51.10
Refined	51.04	31.25	59.38	85.42	78.13	51.05
Random subspace	50.00	23.96	61.46	79.17	76.04	51.00
Coupling	48.96	26.04	61.46	77.08	82.29	51.07
Random directions	60.42	28.13	60.42	84.38	77.08	51.27
Joint	52.08*	29.17*	61.46	79.17	78.13	50.85
Visual	50.00*	38.54*	60.42	81.25	73.96	50.83
Action	42.71*	36.46*	58.33	86.46	76.04	51.11

Table 6: Author-reported benchmark context [3]. Tables 2/11 report late-epoch recipe aggregates; Table 12 reports final checkpoints. Different aggregation, checkpoints and scenarios prevent pairing these rows with our interventions.

Published model	Reach	R.-Wall	Push-T	P.Maze	Wall	DROID
DINO-WM	44.8	35.1	66.0	81.6	64.1	39.4
JEPA-WM recipe, L2	58.2	41.6	70.2	83.9	78.8	48.2
JEPA-WM recipe, L1	55.1	40.8	63.4	79.7	46.7	47.2
JEPA-WM final, L2	49.0	29.2	69.4	83.3	80.9	46.5

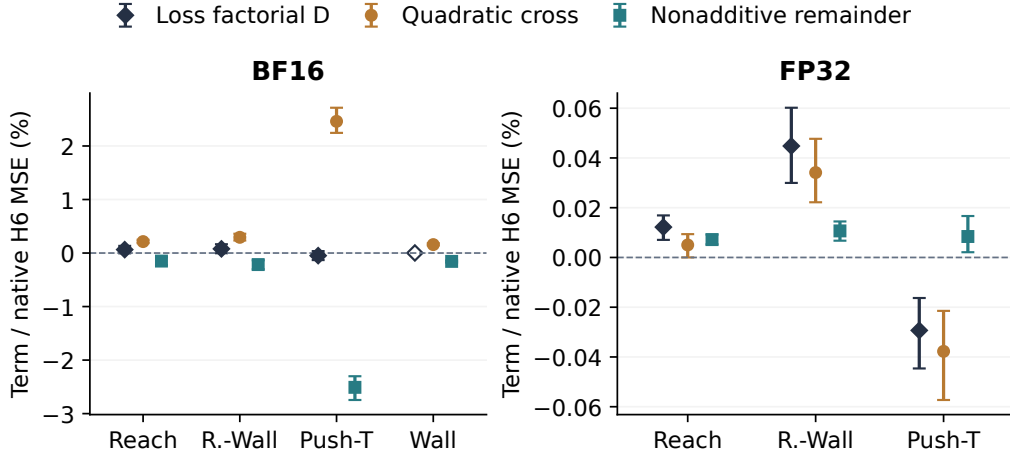
H Local geometry: targets and missingness

Table 7: Reconstruction reduction is $100(L_{\text{linear}} - L_{\text{cubic}})/L_{\text{linear}}$ for omitted-activation MSE. H6 advantage uses the future proprioceptive loss difference divided by *native future MSE*. Positive favors cubic. These denominators are not interchangeable; values are descriptive means.

Task	n	Reconstruction reduction (%)		H6 advantage (%)	
		BF16	FP32	BF16	FP32
Reach	33	-46.4289	94.1933	0.009638	-0.000231
Reach-Wall	27	-46.7716	93.9634	0.026390	0.000000123
Push-T	21	-35.1464	99.9161	-0.021238	0.005017
Wall	192	-43.3160	99.9748	-0.006288	-0.000688
PointMaze	200	-42.2175	99.9858	0.015918	-0.014072

All ten task/precision geometry sources are retained. The seven coupling reports are not a complete five-task-by-two-precision factorial: registered PointMaze coupling and FP32 Wall coupling are absent. Output decomposition is unavailable at H2/H4/H5. The compact Wall source lacks per-arm lineage losses, so its aggregate loss interaction is retained without invented lineage contrasts. Saved interaction norms and cross terms are recounted and identity-checked, not reconstructed by reexecuting output tensors. Precision-specific fits and realized doses prevent isolating rounding as a treatment.

Same-input forecast interaction: $D = \text{cross} + \text{remainder}$



Different y-scales. Remainder already includes output-interaction q^2 .

Wall D: aggregate only, no CI. Other intervals: marginal paired 95%.

No PointMaze or FP32 Wall source. This is not a success factorial.

Precision-specific fits and doses preclude a precision-causal claim.

Figure 11: Output pathways and squared-loss interaction are distinct. All seven registered H6 proprioceptive conditions. The additive cross-term plus nonadditive remainder equals the loss factorial; the remainder already includes $\|q\|^2$, which is not added twice. Wall's aggregate-only factorial has no invented interval; its remaining intervals retain the saved fixed native denominator. Separate axis scales, precision-specific fits, and doses preclude treating the panels as a controlled precision experiment. These are fixed-input forecasts, not protected behavioral interactions.

I Reproducibility and analysis boundaries

The protected output is `reports/fresh-confirmation/report.json`, SHA256 8123d71497835fc164f09f5094c430308647a3ee2462c66746baea32633a0b15. An independent audit checks 32 rates, 384 unique scenario owners, 3,072 arm hashes, and referenced engineering bundles. The registered bootstrap seed is 2026091221. Same-device pairing and scenario-specific RNG hold within each protected scenario, not between old and fresh cohorts. Input-overlap protection is relative to audited project exposure records, not all base-model pretraining.

The decision reanalysis is `paper/data/decision_geometry.json`, its per-scenario CSV, and `analysis/mechanism/decision_geometry.py`. Every source record is hash-bound. Exploratory marginal scenario-bootstrap intervals use 20,000 resamples and seed 2026091321; the finite-bank certificate itself is arithmetic, not statistical.

Layer measurements are in `paper/data/layer_mechanism_grid.csv`. The pathway recount is `paper/data/pathway_geometry.json`, with lineage, summary, and cross-precision exports. Curated baseline provenance and figure captions are in `benchmark_comparison_sources.json` and `benchmark_figure_provenance.json` under `paper/data`. Protected rates are never recovered from pixels or historical denominators.

Execution and scaling. The frozen PyTorch inference workloads used independent whole-scenario GPU replicas, with 300 candidates batched inside each forward call. All eight arms of a scenario stayed on the same physical device. Pinned cloud staging and asynchronous hash-verified preservation overlapped independent execution; they did not alter scientific inputs. Strict FP32 and TF32-off settings were retained. Evaluation uses independent whole-scenario replicas; the model fits on each device and requires no backward pass. Changes to precision, kernels, or parallelism require replay parity checks before scientific comparison.

The code and aggregate package is maintained at https://github.com/stevenybuilder/jepa_steering, currently a private repository. This does not claim that all raw archives are public. New replay or adaptation work requires separately specified inputs, parity checks, and an evidence label. Unfinished experiments are not counted as results here.